Iris Phenolic Liners
Continuous-Heating Campaign

MIT Rocket Team — exploratory numerical study

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Setup and provisional numerical diagnostics.
No validated motor burnthrough time is claimed.

1 New objective and current execution

The November 5 instruction is to choose the exposure duration and answer when the liner is exhausted, rather than repeatedly demonstrating seven-second survival. Four fresh independent radial calculations therefore apply continuous nominal heating until the finite-row model records exhaustion. They are not native ANSYS results. The already-running CF320 ANSYS verification remains intact on ten SMP cores; subsequent short-pulse jobs have been deferred.

The environment retains gas temperature 2504 K (2230.85 C), uncorrected convection coefficient 1000 W/(m^2 K), nominal chamber pressure 800 psi, the documented blowing correction and zero effective hot-side radiation. Initial liner thickness is 4.7625 mm for the sustainer and 6.3500 mm for the booster. These scouts use 160 liner and 40 casing cells and a 0.0015625 s step ceiling. Reporting milestones extend from 30 to 960 s without switching off heating at seven seconds. The chosen environment is hypothetical, not an inferred propellant burn duration.

2 First continuous-heating outcomes

Case	Model exhaustion [s]	Earlier property flag [s]
Sustainer Al	114.9	49.6
Sustainer CF	112.2	18.5
Booster Al	151.7	87.4
Booster CF	147.0	32.9

Table 1: Coarse numerical events of the inherited finite-row closure. All occur after the casing model has left its provisional property range. These are not accepted physical motor lifetimes or allowable firing durations.

The arithmetic now reaches the requested geometric endpoint. It does not yet establish its credibility: neither time/mesh convergence of these events nor high-temperature casing validity has been demonstrated. The prior rejected endurance calculations remain rejected and are not the source of this table. The first 150 C CF or 300 C Al property-range crossing is recorded separately; these are evidence-range flags, not certified structural failure temperatures. Beyond them the equations continue with the inherited surrogate properties.

Source/configuration fingerprints, checkpoint continuity, final-row bookkeeping, monotone sampled time and the cumulative mass ledger pass. Maximum cumulative mass relative error is 1.54e-14, and maximum step

equation-energy residual is 4.2e-11. Closing that discrete energy equation does not prove a physically complete open-system enthalpy model. Every record retains `eligible_for_motor_lifetime=false`.

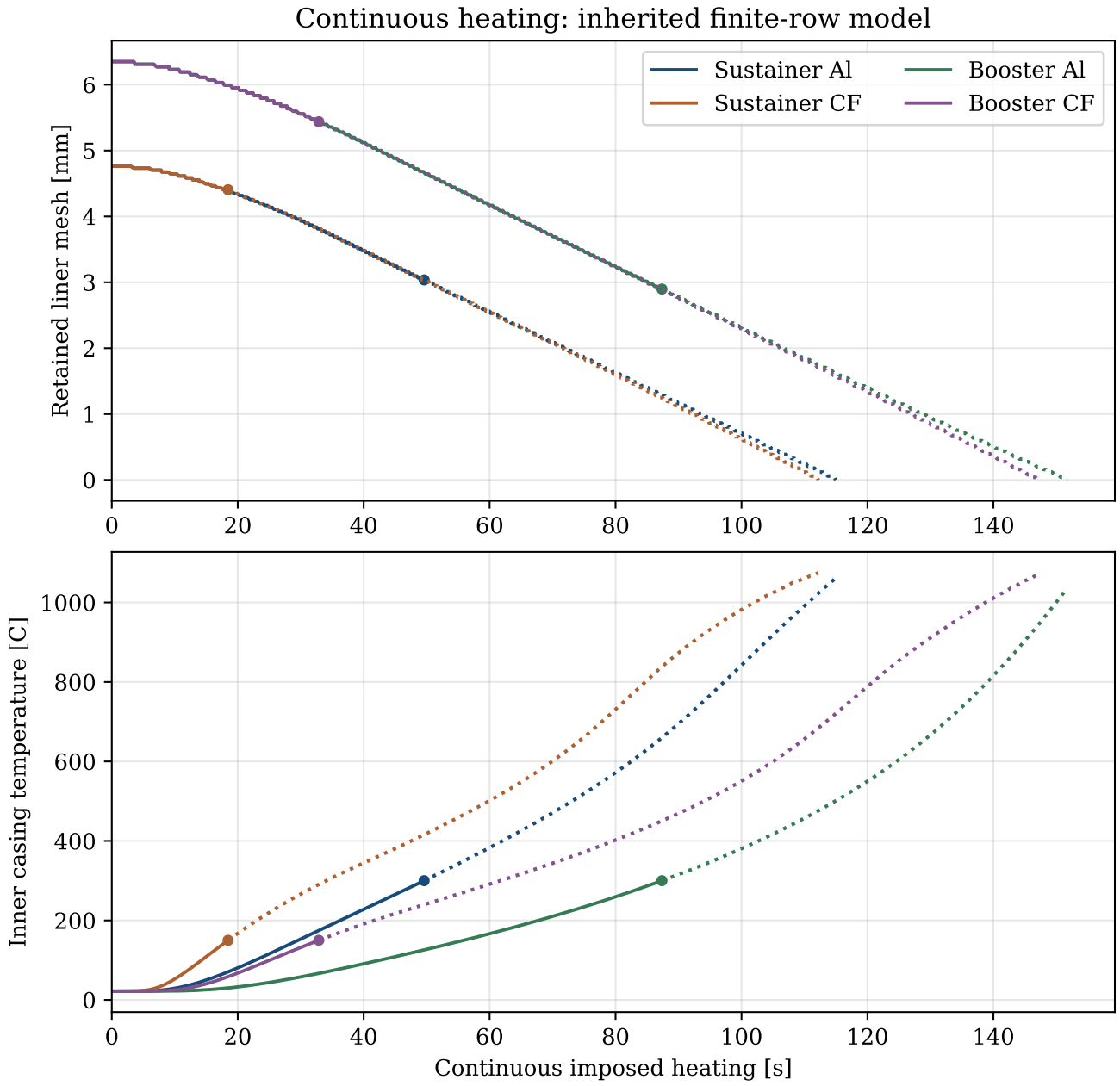


Figure 1: Independent mathematical scouts. Dots mark the first casing-property flag; dotted curves are subsequent surrogate continuation. Remaining thickness is retained mesh, including char, not the thickness of undamaged virgin resin.

3 Numerical checks now prepared

For each motor, a 160/40-cell calculation at half the step isolates temporal error. A subsequent 320/80-cell calculation at that same smaller step isolates spatial change. Those eight calculations are separate from the four scouts; their preparation is not a convergence result. Further 640-cell and time checks follow where required. Event-time change must be below five percent, with intermediate retained-thickness agreement within the larger

of 0.02 mm or one percent of initial thickness. Surface histories need their own assessment.

4 Model improvements and implemented first component

The priority is a conservative moving surface. In the old model, partially consumed rows retain thermal capacity until a residual-mass threshold removes them. This couples the apparent recession to row width and cutoff. The new candidate will update occupied volume, conductive paths and stored enthalpy as the surface moves. The surface velocity must follow a mass/energy closure that distinguishes unreacted material, char and escaping gas.

The geometry/remapping component is implemented and tested separately. In the cylindrical volume coordinate $x = r^2/2$, each conserved field u satisfies

$$I_i = \int_{r_i}^{r_{i+1}} u(r) r \, dr, \qquad \sum I_{\text{old}} = \sum I_{\text{retained}} + I_{\text{swept}}.$$

It remaps extensive mass and enthalpy, not temperature and conversion independently, and returns swept-out quantities explicitly. Analytic nonuniform overlap, zero-motion and prescribed pure-char front tests pass at three resolutions. The artificial 100 s front benchmark is a verification input, not an Iris result. No cell or heat capacity remains after exact domain exhaustion.

This component is not yet a coupled thermal/pyrolysis solver. Next implement thermodynamic reconstruction, reaction/gas enthalpy terms, conservative front transport and an iterated surface energy balance without double-counting removal energy. Then verify a Stefan-type moving-front limit, reaction limits and integrated mass/energy before native adoption. New ANSYS packages will preserve the original sealed runs and restart lineage.

High-temperature Al properties/phase change and CF resin decomposition need documented surrogates; otherwise post-range results remain conditional intact- backing scenarios. Heat-transfer, radiation and char-consumption sensitivities must accompany any eventual event-time range. Measured liner recession and actual exposure history remain the strongest missing physical constraints.

Moving-grid energy, decomposition and surface balances are established elements of charring-ablation modelling; see Chen and Milos (2013), NASA NTRS 20130014432. Its PICA/re-entry parameters are not calibration data for this motor.

A Candidate conservative geometry kernel

Complete source below. This is a new independent component; the earlier full MAPDL/Fortran listings remain in the main report. No native thermal law has been replaced by this kernel yet.

```
"""Conservative radial geometry kernel for a future moving-surface model.

This is NOT a heat-transfer/reaction solver or a recession law. It transports
extensive states under a prescribed boundary movement. Units are per radian
and per metre of axial length; multiply by sector angle and length if needed.
"""
from dataclasses import dataclass
import numpy as np


@dataclass
class Remap:
    density: np.ndarray
    cell_integrals: np.ndarray
    removed_integrals: np.ndarray
    original_integrals: np.ndarray
    balance_residual: np.ndarray


```

```
def advance_front(radius, interface, speed, dt):
    """Locate geometric exhaustion for a prescribed constant normal speed.

    The caller must obtain speed from a verified surface mass/energy balance.
    A temperature, conversion threshold or arbitrary cutoff is not used here.
    """
    if not np.all(np.isfinite([radius, interface, speed, dt])):
        raise ValueError('Nonfinite front inputs')
    if radius <= 0 or interface < radius or speed < 0 or dt <= 0:
        raise ValueError('Invalid front inputs')
    if radius == interface:
        return interface, 0., True
    if speed == 0:
        return radius, dt, False
    to_event = (interface - radius) / speed
    if to_event <= dt:
        return interface, to_event, True
    return radius + speed * dt, dt, False


def remap(old_edges, density, new_edges):
    """Integrate piecewise-constant conserved densities over new radial cells.

    The last dimension of density is the old cell index. Other dimensions can
    hold virgin-equivalent mass, char mass, solid enthalpy and related conserved
    quantities. Remap those quantities, not temperature or alpha independently.
    Thermodynamic reconstruction and reaction/removal source terms are the
    caller's responsibility. Interface positions must be retained by the caller
    so heterogeneous material labels are not averaged into one material.

    New cells cover the retained domain [new inner radius, old outer radius].
    A one-entry new_edges array denotes exact exhaustion of this domain. The
    swept-out mass and enthalpy are returned explicitly, not silently deleted.
    """
    old = np.asarray(old_edges, dtype=float)
    new = np.asarray(new_edges, dtype=float)
    field = np.asarray(density, dtype=float)
    if old.ndim != 1 or new.ndim != 1 or len(old) < 2 or len(new) < 1:
        raise ValueError('Expected radial edge arrays')
    if field.ndim < 1 or field.shape[-1] != len(old) - 1:
        raise ValueError('Density field does not match old cells')
    if not all(np.all(np.isfinite(x)) for x in [old, new, field]):
        raise ValueError('Nonfinite remap inputs')
    if old[0] <= 0 or np.any(np.diff(old) <= 0) or np.any(np.diff(new) <= 0):
        raise ValueError('Cells must have strictly positive radial volume')
    if new[0] < old[0] or new[-1] != old[-1]:
        raise ValueError('Only inner recession with a fixed outer boundary is supported')
    # x = r^2/2 is the exact cylindrical volume coordinate per radian/metre.
    xold = old * old / 2
    xnew = new * new / 2
    old_integrals = field * np.diff(xold)
    cumulative = np.concatenate([np.zeros(field.shape[:-1] + (1,)),
                                 np.cumsum(old_integrals, axis=-1)], axis=-1)
    flat = cumulative.reshape((-1, len(old)))
    retained_cumulative = np.array([np.interp(xnew, xold, row) for row in flat])
    retained_cumulative = retained_cumulative.reshape(field.shape[:-1] + (len(new),))
    integrals = np.diff(retained_cumulative, axis=-1)
    removed = retained_cumulative[..., 0]
    original = cumulative[..., -1]
    residual = original - removed - integrals.sum(axis=-1)
    new_density = integrals / np.diff(xnew)
    return Remap(new_density, integrals, removed, original, residual)
```